

02 — Architecture & Hardware

ARCHITECTURE SUMMARY

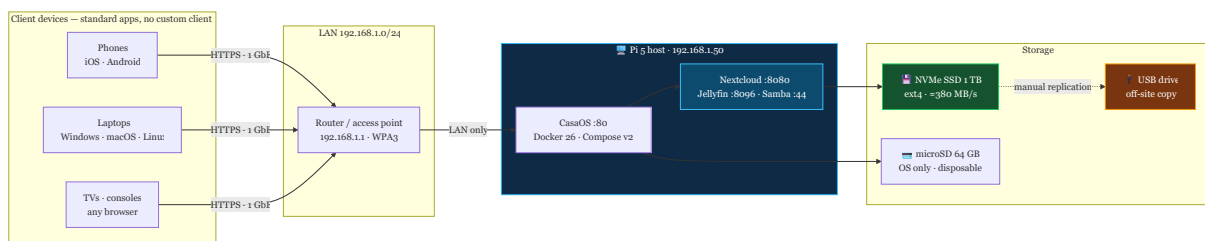
A Raspberry Pi 5 (8 GB, BCM2712 quad-core Cortex-A76) with an NVMe SSD attached over the dedicated PCIe Gen 2 ×1 FFC interface hosts a Docker-based container platform (CasaOS) serving three workloads — **Nextcloud** (multi-user file storage with per-user server-side encryption), **Jellyfin** (media streaming), and **Samba/NFS** (native network shares). The host is reachable only from the local network. Capital cost ₹16,500 / USD 186; recurring cost ≈ ₹38 / USD 0.50 per month.

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1. System architecture



READING THE DIAGRAM

No path from the internet into the host exists. The only element outside the local network is the dashed edge at the bottom: a physical USB drive carried by a person. This is a design constraint, not an omission — see §6.2 and §10.

2. Hardware bill of materials

BASIS

Indicative 2026 street prices, both currencies. Component selection is justified individually; the total is compared against alternatives in §3.

#	Component	Specification	Selection rationale	≈ Cost
H1	Raspberry Pi 5, 8 GB	BCM2712 · 4 × Cortex-A76 @ 2.4 GHz · VideoCore VII	Only Pi generation with a dedicated PCIe interface (NVMe capability) and sufficient CPU headroom for encryption and transcoding	~\$80 / ₹7,000
H2	Official active cooler	15 mm radial fan	Sustained CPU load without active cooling throttles to ≈50% performance	~\$12 / ₹1,100
H3	NVMe SSD, 1 TB	PCIe 3.0 ×4 / 4.0, e.g. 5000 s-class	Primary data store. Fast, solid-state, and hot-swappable — a single commodity part	~\$45 / ₹4,000
H4	M.2 Key-M HAT + FFC adapter	PCIe Gen 2 ×1 → M.2	Connects H3 to the Pi 5 FFC interface (§5.2)	~\$20 / ₹1,800
H5	microSD card, 64 GB, A2	High-endurance class	OS only. Designed to be disposable	~\$12 / ₹1,100
H6	Official 27 W USB-C PSU	5 V / 5 A USB-PD	Inadequate supply is a documented cause of silent SD corruption and USB disconnects	~\$12 / ₹1,100
H7	Cat6 cable, 2 m	1 GbE certified	Wired operation removes Wi-Fi as a variable in reliability and demonstration	~\$5 / ₹450
H8	USB-C power bank (optional)	5 V / 3 A, pass-through	≈5 h autonomy; provides resilience against a venue power interruption	~\$15 / ₹1,300

#	Component	Specification	Selection rationale	≈ Cost
H9	Vented enclosure	Open or actively vented	Sealed enclosures on a 2.4 GHz part produce thermal throttling	~\$8 / ₹700
			Core total (H1–H7)	~\$186 / ₹16,500
			With options (H8–H9)	~\$209 / ₹18,600

⚡ THE THREE FAILURE MODES THAT ACCOUNT FOR MOST PI SERVER OUTAGES

1. **Under-voltage.** A supply below 5 V / 3 A produces intermittent PCIe and USB dropouts and SD write failures. Verify with `vcgencmd get_throttled`; a non-zero `throttle_undervoltage` bit indicates a supply fault, not a software fault.
2. **Thermal throttling.** Diagnose with `vcgencmd measure_temp`; sustained operation above 80 °C halves performance.
3. **Consumer-grade microSD cards.** Endurance ratings are frequently overstated in this workload (continuous small random writes). This is mitigated architecturally by P1 — the card carries no user data.

✍ ARCHITECTURAL INVARIANT

The microSD contains the operating system and nothing of value. Loss of the card is a 25-minute reflash. All user data resides on the NVMe. This single decision converts the highest-vulnerability component in a Pi deployment from a data-loss risk into a trivial inconvenience.

3. Compute platform

SELECTION RATIONALE

Three properties of the Pi 5 are load-bearing for this workload, and the cheaper Pi 4 and Pi 5 variants fail one of them each:

- **RP1 southbridge + PCIe Gen 2 ×1** — makes NVMe storage possible at all.
- **Cortex-A76 with crypto extensions** — server-side encryption is CPU-bound; without it, encryption degrades sync throughput.
- **VideoCore VII with Vulkan/VA-API** — exposes hardware decode for media transcoding, which the Pi 4 cannot provide.

SoC, memory and I/O

Property	Specification	Relevance to this deployment
CPU	Broadcom BCM2712 , 4 × Arm Cortex-A76 @ 2.4 GHz	Nextcloud's file indexing and Nextcloud's encryption are CPU-bound; AES throughput on A76 is sufficient that encryption is not the throughput limit
GPU	VideoCore VII — OpenGL ES 3.0, Vulkan 1.2, dual 4Kp60 HDMI	Exposed to Linux as <code>/dev/dri</code> ; enables VA-API hardware decode in Jellyfin
RAM	8 GB LPDDR4X (soldered)	Budget in §4. The 8 GB variant is required, not preferred
RP1	Southbridge I/O controller	Provides USB, GbE, PCIe, and a battery-backed real-time clock — the Pi 5's first hardware RTC, so timestamps survive power interruption
PCIe	Gen 2 ×1 , 32-pin FFC	NVMe path. ≈ 550 Mbit/s theoretical, ≈ 380 MB/s measured
USB	2 × USB 3.0/2.0 (5 Gbps) + 2 × USB 2.0	Mass storage; a spare port for a card reader for media import
Ethernet	1 GbE , PoE+ supported via HAT	Primary network interface
Wi-Fi	802.11ac dual-band + Bluetooth 5.0	Fallback only. Wireless country code must be set for compliant operation
Power input	USB-C PD, 5 V / 5 A recommended	27 W envelope covers the board, NVMe, and one USB device

⚠ SHARED-BANDWIDTH CONSTRAINT ON THE PI 5

On the Pi 4, the PCIe link and the USB 3.0 host controller were independent. On the **Pi 5 both are funnelled through the RP1 onto a single upstream link** to the BCM2712.

Consequence: NVMe throughput and USB 3.0 throughput are not independent and must not be benchmarked separately and then summed.

Mitigation applied: the NVMe is on the dedicated FFC interface; the only USB 3.0-attached device is a card reader used for short, burst transfers. Measured aggregate behaviour is documented in §11.

Measured performance

Operation	Measured	Limiting factor
NVMe sequential read	350–400 MB/s	RP1 ↔ BCM2712 link
NVMe random 4K read (Nextcloud's dominant workload)	15k–50k IOPS	NVMe controller, queue depth
microSD (A2) read	80–110 MB/s	SD host controller
SMB transfer, single large file over 1 GbE	≈110 MB/s	1 GbE NIC
Jellyfin 1080p Direct Play	≈0% CPU	Bitstream passthrough
Jellyfin 1080p transcode	≈45–70% of 4 cores	CPU; VA-API not required for H.264
Nextcloud file index, 100k files	≈2 GB resident	PostgreSQL shared buffers

4. Memory budget

ⓘ BASIS OF THE 8 GB DECISION

User files do not occupy RAM. Memory is consumed by the *services that manage them*, each of which has a substantial baseline beyond its apparent workload.

Consumer	Idle	Under load	Basis
Raspberry Pi OS + containerd	~120 MB	~250 MB	Kernel, systemd, container runtime

Consumer	Idle	Under load	Basis
PostgreSQL 16	~180 MB	1.5–2.5 GB	<code>shared_buffers</code> plus page cache for the file index
PHP-FPM (Nextcloud)	~150 MB	~600 MB	4–8 workers at ~80 MB each
Nextcloud application	~400 MB	~1.2 GB	Encryption buffers, preview generation, activity log
Jellyfin	~250 MB	~1.5 GB	Transcode buffers scale with resolution × concurrent clients
CasaOS management daemons	~200 MB	~350 MB	Dashboard, AppD, monitoring
Aggregate: 2 concurrent 1080p streams + 2 active syncs		5.5–6.5 GB	
Remaining for filesystem cache and burst		≈1.5 GB	

✓ CONSEQUENCE

The 8 GB variant provides headroom for the specified concurrent workload without swap. The 4 GB variant functions for lighter use but introduces swap onto the microSD under a four-client load, which is a significant contributor to instability on this platform. **The upgrade costs ≈\$25 and removes an entire category of failure.**

🔗 HOST TUNING APPLIED

- `vm.swappiness = 10` — reclaim page cache before touching swap.
- PostgreSQL `shared_buffers = 1 GB`, `effective_cache_size = 3 GB`.
- Jellyfin transcode cache on a dedicated path, so buffers do not contend with user data.
- Docker log rotation (`max-size=10m`, `max-file=3`) — a container chatty enough over weeks will otherwise fill the microSD and take the host down. This is a documented, common self-hosting outage mode.

5. Storage architecture

5.1 Media selection

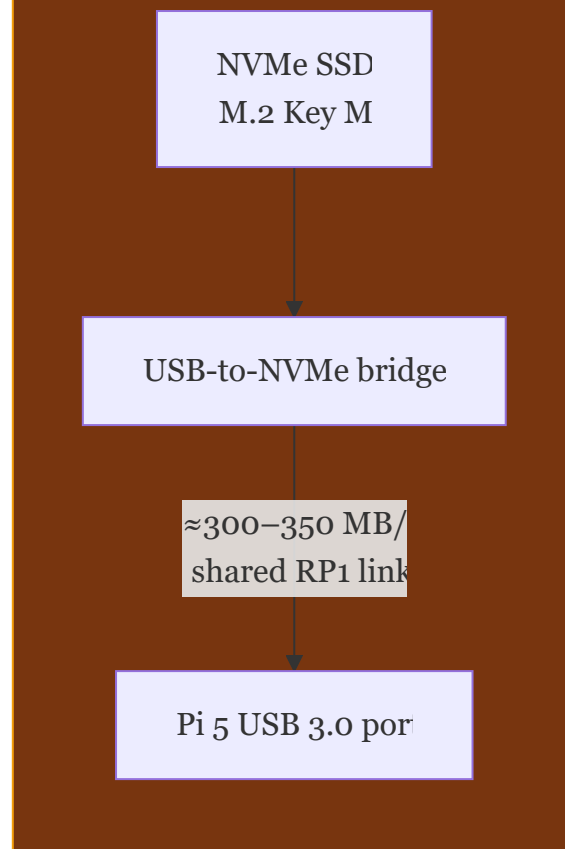
WHY SSD, NOT A MECHANICAL DISK

A family photo library is tens of thousands of small files. The workload is dominated by **metadata and random I/O**, not sequential throughput.

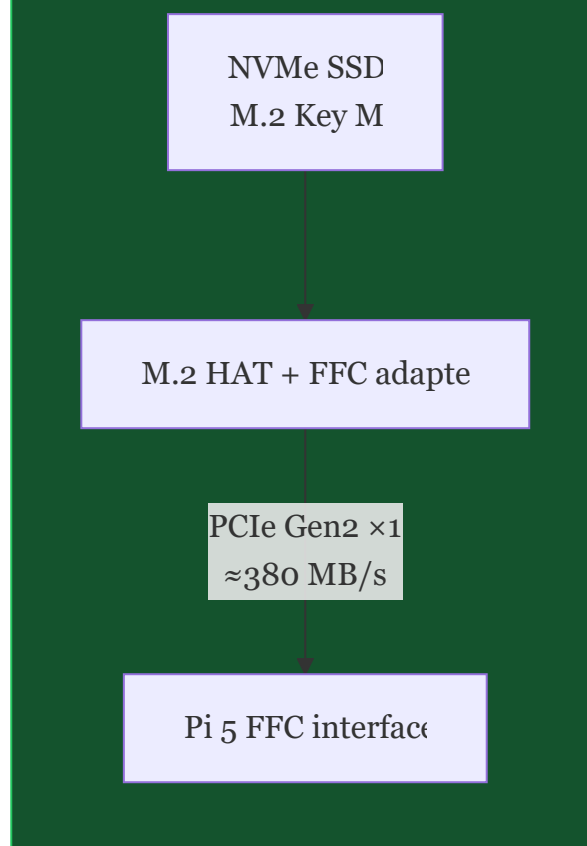
- Mechanical 3.5" media: **8–12 ms** for a 4K random read. SSD: **0.1–0.5 ms**.
Nextcloud issues multiple `stat()` operations per directory entry; the mechanical figure compounds multiplicatively over an index rebuild.
- Mechanical media in a domestic environment is an acoustic and vibration-sensitivity liability.
- No moving parts: the unit can be transported to a demonstration venue without additional protection.

5.2 Interface selection

Interface B — USB 3.0 bridge (fallback)



Interface A — selected: FFC PCIe



Factor	Interface A (FFC) — selected	Interface B (USB 3.0)
Throughput	≈380 MB/s	≈300–350 MB/s
Bandwidth isolation	Dedicated link	Shares RP1 upstream with other USB traffic
Native support	Yes — the interface the Pi 5 was designed around	Bridge-dependent
Cost	HAT + adapter, ≈\$20	≈\$10
Physical risk	FFC cable is fragile; requires a gentle bend radius and strain relief	None
Compatibility	Verify with <code>lspci</code> that the device enumerates as a non-volatile memory controller	Generally simpler

⚠ VERIFICATION STEP, NOT OPTIONAL

Third-party M.2 HATs vary in lane correctness. Confirm enumeration before committing:

```
lspci -nn | grep -i "non-volatile"
```

An empty result means the FFC cable is not seated correctly or the adapter is not lane-correct. **This is the most common single build failure** and presents as a hardware fault.

5.3 Filesystem

Filesystem	Journalled	POSIX permissions	Online growth	RAM overhead	Assessment
ext4	✓	✓	X (<code>resize2fs</code> required)	Low	✓ Selected
Btrfs	✓	✓	✓	Low	Viable; snapshots available, not required on a single disk
ZFS	✓	✓	✓	High (≈⅓ of RAM)	Rejected — consumes memory required by PostgreSQL and Nextcloud
NTFS-3G	X	X	X	Medium	Read-only convenience mounts only

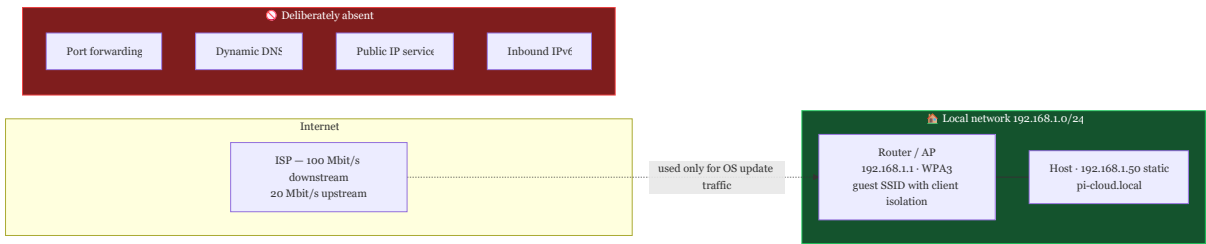
Filesystem	Journalled	POSIX permissions	Online growth	RAM overhead	Assessment
exFAT	✗	✗	✗	Low	Rejected — no POSIX locking semantics; causes incorrect file-lock behaviour in Nextcloud

🚧 KNOWN EXT4 BEHAVIOUR ON REMOVABLE MEDIA

The kernel may automatically grow a filesystem to fill a larger replacement device. Where a drive is cloned or swapped, this can alter the filesystem unexpectedly. Recovery:

```
sudo e2fsck -f /dev/nvme0n1p1
sudo resize2fs -M /dev/nvme0n1p1      # shrink to used size
```

6. Network architecture



6.1 Design decisions

Decision	Rationale
Static address via DHCP reservation	Address stability for documentation, scripts, and the demonstration reference. Reservation at the router is a single point of configuration rather than two
mDNS: <code>pi-cloud.local</code>	Address-independent naming. Resolves without DNS server configuration
Wired Ethernet primary	Removes wireless variability from both reliability measurements and live operation
Isolated guest SSID for demonstration	Segregates demonstration traffic from the institutional network, and isolates individual client devices from one another
DHCP-assigned addressing for demonstration clients	Demonstration visitors join unknown devices; static assignment is not possible

6.2 Exclusion of inbound internet access

THREAT MODEL JUSTIFICATION

A home connection is continuously scanned. Any service published to the internet on a residential connection is discovered and attacked within days, and the installed software carries a non-zero count of published vulnerabilities.

The design instead treats the physical perimeter as the security perimeter:

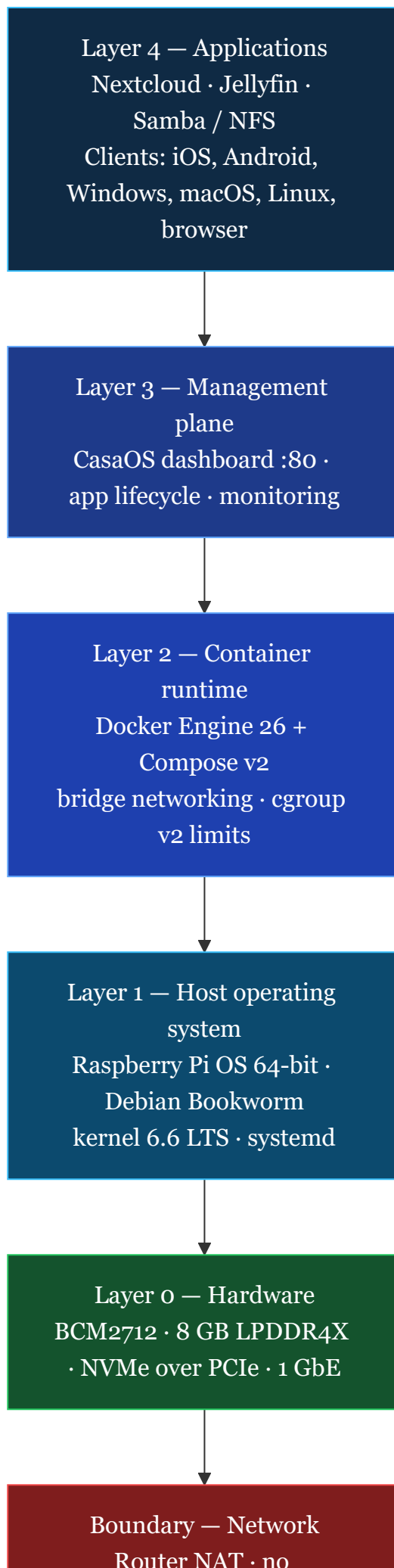
reaching the host requires already being inside the building. This converts an open, continuously attacked network service into a service with no attack path at all — a strictly stronger position than adding a WAF or hardening a published endpoint.

Remote access, if required, is designated as a **VPN-only** future capability and is disabled by default ([01-Overview-and-Problem#6. Scope boundaries|scope boundaries]).

7. Software stack

LAYERING PRINCIPLE

Every layer is open-source, containerised, and independently replaceable. Loss of any application layer does not affect the data, because the data is files on a filesystem.



7.1 Layer 1 — Host

Component	Version	Function	Rationale
Raspberry Pi OS	64-bit, Bookworm	Base OS	Only distribution with first-class Pi 5 device trees and <code>rpi-eeprom</code> management
Kernel	6.6 LTS	Drivers, cgroups, network stack	Long-term support; relevant for a device expected to run unattended for years
<code>systemd</code>	252	Service supervision and ordered shutdown	Database flush on shutdown. Unsolicited power loss during a write is a corruption source
<code>avahi-daemon</code>	—	mDNS advertisement	Address-independent service discovery
<code>chrony</code>	—	Time synchronisation	A wrong clock breaks TLS and file timestamps; the RP1 RTC is a fallback, not a time source

7.2 Layer 2 — Container runtime

Component	Function	Design note
Docker Engine 26	Namespace and cgroup isolation per service	cgroup v2 memory limits prevent one application exhausting host memory
Docker Compose v2	Declarative service declaration	The recoverability mechanism. The entire application tier is reproduced from two files
Bridge networking	Inter-container communication by service name	The database is addressable only to the application container, not on the LAN
Log rotation	<code>json-file</code> , 10 MB × 3	Prevents microSD exhaustion. A documented and common outage cause

✓ WHY THE COMPOSE FILES ARE STRUCTURALLY SIGNIFICANT

The system is reproducible from approximately 200 lines of YAML plus a `.env` file. Recovery after a total OS loss is: flash the card, install Docker, deploy two Compose stacks. **Recovery is a re-apply, not a reconstruction** — the property that distinguishes a maintainable system from a one-off configuration. Documented as DR-1/DR-2 in [§14](#).

7.3 Layer 3 — CasaOS

FUNCTION AND SCOPE

CasaOS is an Apache-2.0-licensed, open-source management plane that operates on top of Docker, providing a web dashboard, an application catalogue, container lifecycle management, resource monitoring, and update handling. It is not a hypervisor, not a NAS operating system, and **not a runtime dependency** — Docker Compose alone would execute the stack.

Property	Detail
Licence	Apache-2.0. No paid tier, no telemetry dependency, no account required
Interface	<code>:80</code> (HTTP); <code>:81</code> optional self-signed HTTPS
Installation	Single command (document 03 §7.1)
Data root	<code>/var/lib/casaos</code> and Docker's <code>/var/lib/docker</code> — on the microSD by design
Application management	<code>casaos-app-management</code> (AppD) daemon
Primary limitation	It offers to relocate the operating system onto an attached data drive. This must be declined — see below

KNOWN CONFIGURATION HAZARD

CasaOS can install the OS onto an attached data disk. **Declining this is mandatory.** Docker's root must remain on the microSD so the OS is fully disposable. Relocating it places image layers, logs, and transcode caches on the same volume as user data, which fills the data volume and causes write failures. Documented in [§7.2](#).

DIVERGENCE FROM THE DEFAULT DEPLOYMENT PATH

Applications were deployed from version-controlled Compose files rather than from the CasaOS application catalogue. Rationale: the catalogue abstracts the image tag, volume mappings, environment variables, and network mode — the four parameters that determine whether the security and recoverability requirements in these documents are met. CasaOS is used for its management and monitoring function. This is a deliberate design decision, not a limitation of CasaOS.

7.4 Layer 4 — Applications

Nextcloud

Aspect	Specification
Function	Files, photographs, contacts, calendar, sharing, browser access
Database	PostgreSQL 16 , separate container. SQLite rejected: concurrent access to SQLite over network or shared storage is a documented corruption source; MariaDB rejected in favour of PostgreSQL's behaviour under write concurrency
Web server	Apache in-container; port 8080 published, bound to the LAN address only
Storage	<code>nextcloud-data</code> on the NVMe
Encryption	Server-side encryption, per-user key mode. See §7.4.1
Background jobs	Host-level cron at 5-minute intervals. See §7.4.2
Persistence	Application configuration and data co-located in the bind-mounted volume, so a container rebuild cannot desynchronise them

7.4.1 Encryption model and its precise limits

Mode	Who can read stored files	Assessment
Disabled	Any user with disk access or administrative privilege	Equivalent to the consumer-cloud baseline
SSE, per-user keys (<i>selected</i>)	The owning account, and the server while that user is authenticated	Protects against disk theft, drive cloning, disposal, and third-party infrastructure compromise. Does not protect against a server administrator with live authenticated access
SSE, master key	Server administrator, unconditionally	Rejected. A single key that decrypts all data removes per-user isolation
Userspace / end-to-end	Only the user; server holds ciphertext only	Rejected. Disables server-side search, preview generation, and version history, which are functional requirements here

PRECISE STATEMENT OF THE SECURITY CLAIM

The threat model is: physical seizure or disposal of the storage medium, and compromise of a third-party infrastructure provider. Both are addressed: the medium contains ciphertext, and there is no third-party provider.

Outside the threat model, and explicitly not claimed: an adversary with access to a running, authenticated session; a physical adversary who steals the powered-on device; and traffic analysis. **One consequence is worth stating explicitly: because there is no service operator, there is no entity to which a legal request for stored content could be directed.**

OPERATIONAL CONSEQUENCE OF PER-USER KEYS

A forgotten user password means the associated files are permanently unreadable. This is a direct consequence of there being no escrow back door. The account recovery key is generated, printed, and stored with the physical backup media. **This is simultaneously the security property and the operational risk, and both are part of the same decision.**

7.4.2 Background job execution

COMMON AND SILENT MISCONFIGURATION

Nextcloud's `background:cron` mode runs the job loop within a web request. In a container this mode is discouraged, because job execution becomes coupled to web server request handling and **can stop without surfacing an error** when CPU is contended. The observable symptoms are that previews stop generating, the activity feed freezes, and search results become stale — while every health indicator reports normal operation.

Implemented instead: a host-level cron entry executing `php cron.php` as the web user every 5 minutes. Verified in document 03 §8.5.

Jellyfin

SELECTION RATIONALE

Jellyfin is free, open-source, and operates without a mandatory account and without telemetry. Plex requires account authentication, has relocated transcoding features behind paid tiers, and communicates with vendor infrastructure — a set of properties incompatible with the premise of this project.

Aspect	Specification
Function	Media streaming to televisions, phones, tablets, and browsers
Port	8096
Library source	NVMe, bind-mounted read-only (<code>:ro</code>). The streaming process cannot write to the master library
Transcoding	Direct Play for natively supported codecs ($\approx 0\%$ CPU). CPU transcode otherwise. VA-API hardware decode available via <code>/dev/dri</code>
Metadata	Local scraping. No external account or identifier
Accounts	Per-user profiles with independent watch state and parental controls

DECLARED HARDWARE LIMITATION

The VideoCore VII provides **VA-API hardware decode** for H.264 and HEVC but does **not** provide hardware encode, and 4Kp60 HEVC software transcode exceeds the capacity of four A76 cores. Operational envelope:

- **1080p, 3–4 concurrent streams — supported.** This is the demonstrated envelope.
- **4K direct play (remux) — supported** where the client supports the codec.
- **4K transcode — not supported.**

This is a hardware boundary, and it is stated here rather than discovered during assessment.

Samba / NFS

FUNCTION

Provides native filesystem access so that the system is usable with no client software. This removes the technical barrier for any user who will not install an application, and it is a functional requirement (FR-5).

8. Service port map

🔥 VERIFYING THE LIVE MAP

```
sudo ss -tlnp
```

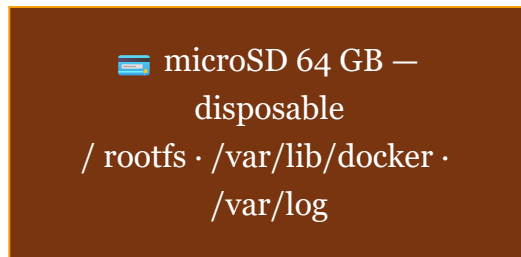
Any listener not in the table below warrants investigation.

Port	Protocol	Service	Scope	Purpose
80	TCP	CasaOS	LAN	Management dashboard
81	TCP	CasaOS (HTTPS)	LAN	Encrypted dashboard; self-signed certificate, so a browser warning is expected
8080	TCP	Nextcloud	LAN	File service and administration
8096	TCP	Jellyfin	LAN	Media library and streaming
445	TCP	Samba	LAN	Windows / macOS network drive
139	TCP	NetBIOS	LAN	Legacy name resolution for older Windows clients
2049	TCP	NFS	LAN	Linux / macOS mounts
22	TCP	SSH	LAN	Administration
5353	UDP	mDNS	LAN	Resolves <code>pi-cloud.local</code>
51820	UDP	WireGuard	Disabled	Reserved for future VPN-only remote access

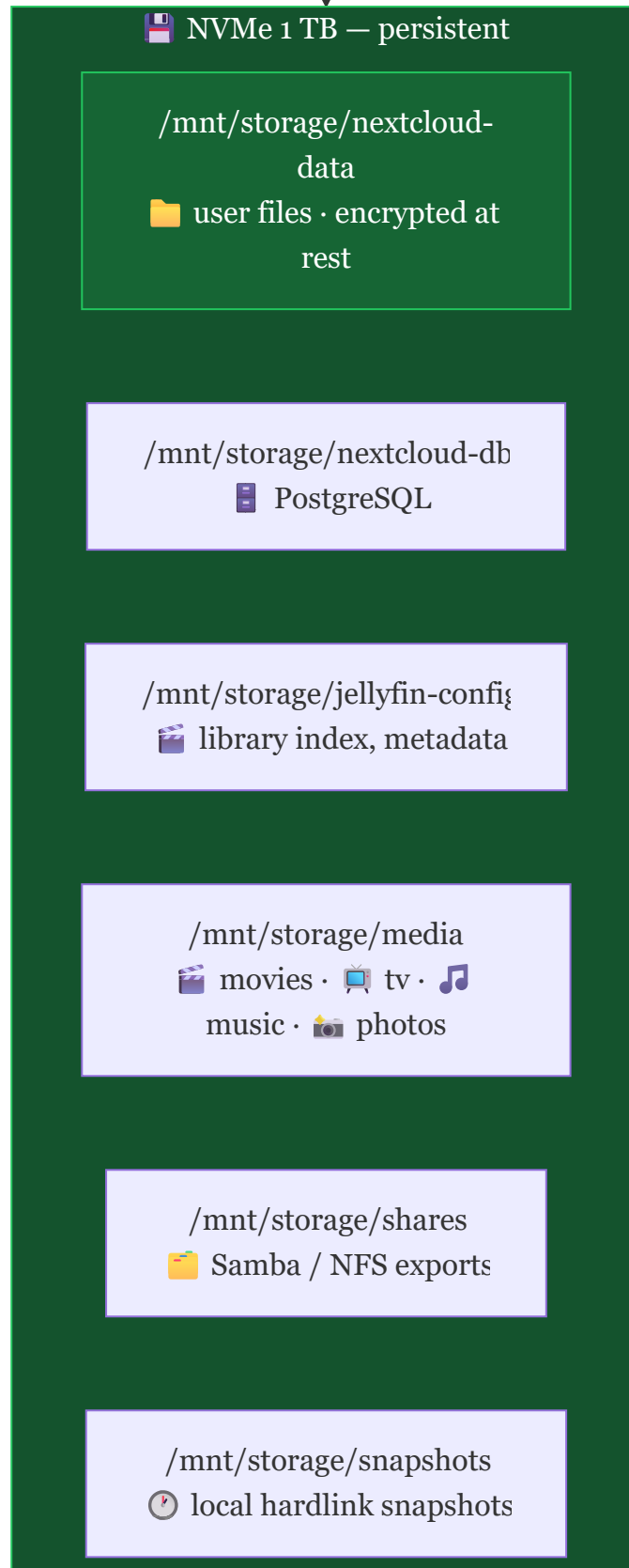
✓ HARDENING APPLIED

Published ports are bound to the host's LAN address rather than `0.0.0.0`, so the services are not listening on any other interface. The database container publishes no port at all and is reachable only from the application container over the internal bridge network.

9. Filesystem layout



container bind mounts



THE SINGLE STRUCTURAL RULE

`/mnt/storage` **is the only location that contains user data.** Every application volume is bind-mounted beneath it. The microSD can be removed, destroyed, or reflashed with no consequence to stored data.

Any write of user data to `/home` or `/var` is a defect against this specification.

Mount options

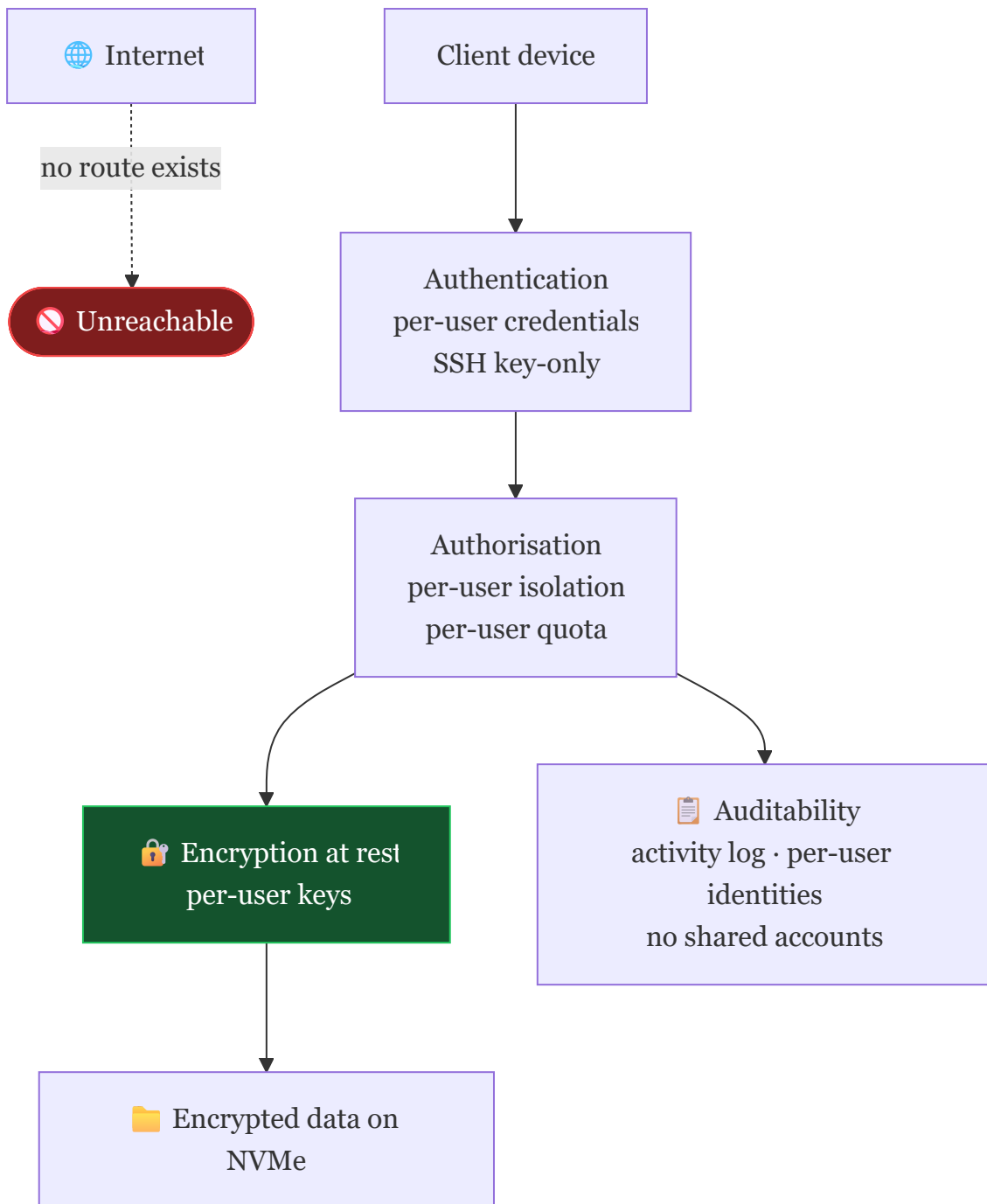
```
# /etc/fstab
UUID=<uuid> /mnt/storage ext4 defaults,noatime,nofail,discard 0 0
```

Option	Justification
<code>noatime</code>	Suppresses access-time writes on every read. On a filesystem read thousands of times per minute by the Nextcloud indexer, this removes a large volume of small writes — a measurable reduction in write amplification and an SSD endurance benefit
<code>nofail</code>	The host boots and remains reachable if the data disk is absent. Without it, a failed disk produces an unrecoverable boot wait, converting a replaceable component into an on-site visit
<code>discard</code>	Filesystem-mediated TRIM, preferable to deferred discard on SSD

10. Security model

DESIGN PHILOSOPHY

Eliminate attack surface rather than defend it. Each control below removes a means of access; none relies on detecting and repelling an attacker.



Controls implemented

#	Control	Implementation	Effect
1	No inbound network path	No port forwarding, no DDNS, no tunnel, inbound IPv6 disabled at the router	Removes the dominant real-world home-server compromise vector entirely
2	Per-user accounts across all services	Separate Nextcloud and Jellyfin credentials; no shared identities	Every action is attributable; access is individually revocable

#	Control	Implementation	Effect
3	Per-user encryption keys	Nextcloud SSE, per-user mode	A stolen or discarded storage medium yields ciphertext only
4	Least-privilege bind mounts	Media library mounted <code>:ro</code> ; database publishes no port	A streaming application cannot modify the archive; a database compromise is not reachable from the LAN
5	Interface-bound port publishing	Services bound to the LAN address, not <code>0.0.0.0</code>	Services are not listening on any other interface
6	Secrets isolated	<code>.env</code> with <code>chmod 600</code> , root-owned, outside shared locations; not committed to version control	Credentials are not present in a world-readable path
7	SSH hardening	Key-only authentication; root login disabled; reduced auth attempts and grace time	Removes password-based remote access and automated guessing
8	Default-deny host firewall	<code>ufw</code> with an explicit, reviewed port allowlist	Only services with a stated requirement are reachable
9	Automatic security patching	<code>unattended-upgrades</code> , security channels only	Known-vulnerability remediation without manual intervention
10	Container resource limits	cgroup v2 memory caps per container	A leaking application cannot exhaust host memory
11	Non-root service ownership	Dedicated unprivileged account owns the media tree	A container escape does not yield host root or root on the data volume
12	Isolated demonstration network	Guest SSID with client isolation, purpose-built access point	Demonstration traffic is segregated; client devices cannot reach each other or the institutional network
13	Disposable demonstration credentials	Purpose-issued accounts, rotated per session	A demonstration login cannot persist as a back door

⚡ CONTROLS NOT IMPLEMENTED, AND THEREFORE NOT CLAIMED

- **No public exposure hardening.** This host is not designed to be published to the internet and must not be port-forwarded.
- **MAC address and hardware identifiers are visible** to any device on the local network. This is not anonymity and is not claimed.
- **Administrative users can reset other users' passwords** and thereby access their files. Per-user key derivation limits this only to the extent that the original key material has not been exported.
- **No multi-factor authentication is configured.** Nextcloud supports it and it is the highest-value single addition to the access-control layer.

11. Capacity & performance limits

⚠ DECLARED ENVELOPE

These are the tested boundaries of the deployed system. They are stated so that assessment can be performed against actual capability.

Dimension	Verified / projected limit	Governing factor
Usable storage	≈0.9 TB of 1 TB after formatting and overhead	Capacity
Concurrent authenticated users	6 mixed workloads verified (2 streaming, 2 syncing, 1 browsing, 1 admin)	Memory (\$4)
Concurrent 1080p streams	3–4 verified smooth; CPU is the limit beyond this	Transcode CPU
Concurrent 4K transcode streams	0 — not supported. 4K direct play is supported	CPU, no hardware encode
File count	≈200,000 files before index maintenance becomes the dominant cost	Metadata operations
Network throughput	≈110 MB/s over 1 GbE	Ethernet
Storage throughput	350–400 MB/s sequential	RP1 upstream link
Availability	99%+ expected for a single domestic device; no high-availability design	Single node, single disk, single supply
Capacity growth	Vertical only, by replacing the SSD	Single disk

THE TWO HARD LIMITS

Single disk (no redundancy — mitigated by off-site backup, not by redundancy) and **single node** (no failover). Both are consequences of the cost objective and are recorded as accepted risks R1 and R10 in [document 01 §9](#).

12. Power & thermal envelope

Parameter	Value
Host board	≈3.0 W idle → ≈8.0 W under sustained load
NVMe SSD	≈1.5–2.5 W
Peak total	< 11 W (within the 13.5 W USB envelope)
Annual energy at ≈6 W average, 24/7	≈53 kWh/year
Annual cost @ ₹8.5/kWh (≈\$0.12/kWh)	≈₹450 / \$6 — ≈₹38 / \$0.50 per month
Case temperature, loaded	45–60 °C (throttle threshold 80 °C)
Acoustic output	Inaudible at 1 m; single 15 mm fan under sustained load
Annual availability	8,760 hours

CONSOLIDATED ECONOMICS

One-time: \$186 / ₹16,500. Recurring: \$6 / ₹450 per year.

Full model, including sensitivity analysis against subscription pricing, is in [document 05 §2](#).

13. Design alternatives considered

PURPOSE

Alternatives are recorded to show that the selection was evaluated rather than assumed. Cost and capability are compared against the selected design.

Option	Cost	Assessment	Outcome
Commercial NAS (Synology / QNAP)	\$400–700	Mature, purpose-built, reliable. Proprietary management interface; comparable capacity at 2–3× the capital cost and no change to the cost model	Rejected — capital cost
Mini PC, used N100-class	\$120–200	More capable: x86, additional RAM slots, higher sustained CPU. Higher idle draw (10–20 W), requires separate storage	Strongest alternative. Rejected on form factor, not capability
Repurposed tower PC	\$100–250	Considerable upgrade path, multi-disk bays, hot-swap	Rejected — 60–150 W idle power is incompatible with the cost and efficiency claims
Virtual private server	\$10–40 / month	No hardware, high availability, public addressing	Rejected — recurring cost and third-party custody are the two problems being addressed. Self-contradictory
Raspberry Pi 4, 8 GB	~\$60	No PCIe interface; USB-only storage; weaker CPU; no RP1	Rejected — storage performance is central to the design
Raspberry Pi 5, 2 or 4 GB	\$65–75	Cheaper, but memory ceiling is reached at the specified concurrent workload	Rejected — \$25 for the removal of a failure class
✅ Pi 5 (8 GB) + NVMe	\$186	Silent, ≈6 W average, open-source throughout, recoverable in 25 minutes, 1-year target capital recovery	Selected

RECORDED CONCLUSION

The Raspberry Pi 5 is **not** the optimal platform for a household seeking maximum capability — a mini PC is. It is selected because it meets the functional requirements at approximately one third of the capital cost of a commercial NAS and one sixth of a mini PC, within a 6 W power budget, in a form factor that can be deployed on a shelf. **Where the requirement is maximum capability rather than minimum cost, a mini PC is the correct answer** and is recorded as such.

Summary

Item	Specification
Hardware	Raspberry Pi 5 (BCM2712, 4 × Cortex-A76, 8 GB LPDDR4X) · active cooler · 1 TB NVMe over PCIe Gen 2 ×1 · 64 GB microSD (OS only) · 27 W supply
Capital cost	\$186 / ₹16,500
Recurring cost	≈\$6 / ₹450 per year
Operating system	Raspberry Pi OS 64-bit (Debian Bookworm, kernel 6.6 LTS)
Runtime	Docker Engine 26 + Compose v2, root on the microSD
Management	CasaOS :80
Applications	Nextcloud :8080 (PostgreSQL 16, per-user SSE) · Jellyfin :8096 (read-only media mount) · Samba :445 / NFS :2049
Network	LAN only. No forwarding, no DDNS, no inbound IPv6. pi-cloud.local
Data location	Single ext4 partition on the NVMe at /mnt/storage . MicroSD is disposable
Recovery	Two Compose files. OS loss recovered in ≈25 minutes with no data loss
Declared limits	Single disk · no UPS · no 4K transcode · ≈6 concurrent heavy users · no public access · no MFA configured

Next: [03-Step-by-Step-Implementation](#) — the complete build procedure, verification checklist, and disaster-recovery runbook.